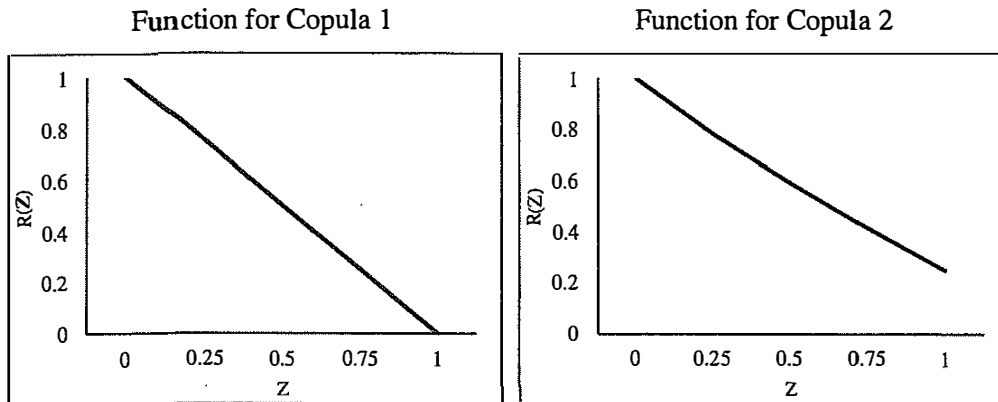


25. (1.25 points)

An insurer writes commercial property and workers compensation insurance in the same geographical area. The insurer is considering two copulas to model the joint distribution of these two lines of business.

Below is a plot of the right-tail concentration function for each copula:



a. (0.75 point)

Discuss which function is more appropriate if the insurer is writing business in an area with high exposure to earthquakes.

b. (0.5 point)

Briefly explain why graphs of right-tail concentration functions can often be misleading and recommend a solution to this problem.

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QUESTION 25	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE: C6: Evaluate and select appropriate models to handle diverse risks, including stochastic approaches.
SAMPLE ANSWERS (BY PART, AS APPLICABLE)	
Part a: 0.75 point	
<i>Sample Answer 1</i> Copula 2 because earthquakes can cause losses for both commercial property and workers compensation (so they are correlated). The $R(z)$ for copula 2 shows that there is correlation between the lines because $R(z) > 0$ as $z \rightarrow 1$. <i>Sample Answer 2</i> Copula 2 is better since $R(z)$ as $z \rightarrow 1$ is greater than 0. In an adverse scenario like earthquake both workers compensation and commercial property will suffer loss. People may seek out of work pay and properties are destroyed. During normal time, the two may not be as correlated	
Part b: 0.5 point	
<i>Sample Answer 1</i> A copula can have significant tail dependence even if $R(1) = 0$, as the function could decrease rapidly. A solution is to look at the function at values a bit below 1 and assess the strength of the dependence. <i>Sample Answer 2</i> It only shows you the right tail of the copula. It should be combined with the left tail concentration function which will show what to expect in the left tail. We focus on $L(z)$ for $0 \leq z \leq .5$ and focus on $R(z)$ for $.5 \leq z \leq 1$. <i>Sample Answer 3</i> There may be sparse data in the right tail which makes it volatile and hard to predict. A solution would be using industry data as reference. <i>Sample Answer 4</i> It is a one-dimensional representation of a 2-dimensional correlation, so can be misleading. View a 3D graph of the joint distribution $C(u,v)$ instead. <i>Sample Answer 5</i> Depending on the copula used, you can see many different right tails and it's hard to	

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visually see which one fits best. Checking the tau of the copulas will give you a better idea of which copula best fits the data.
EXAMINER'S REPORT
Candidates generally scored better on part a. than on part b. Candidates also almost always had a response for part a but often gave no response for part b. Part b. is worded rather broadly, allowing for many different valid responses. Most candidates had a fairly good idea of what the correct or expected answer for part a was but there were many different types of answers for part b.
Part a
Candidates were expected to evaluate two different copulas in the presence of correlation and right tail dependency. Candidates had to: - Identify the correct copula. - Describe why the selected copula was correct by describing the function/graph. - Explain why there is correlation in this particular situation (earthquake causes two usually uncorrelated lines to have correlated losses in the tail). Candidates almost always identified the correct copula. However, many candidates did not explain why the selected graph actually indicated a correlation in the right tail. Another common error was not identifying that the two lines of business were correlated because of the earthquake exposure.
Part b
Candidates were expected to identify problems with right tail concentration graphs and to offer a solution to the problem(s) they identified. Candidates had to identify one problem with right tail concentration graphs and offer a solution to that that was tied to this particular problem. The most common error was to identify problems that were too vague or ambiguous. Also, quite often, candidates offered a solution that didn't match the issue they described. Finally, sometimes candidates offered a problem but no solution and vice versa. Because of the open way the question was asked, there were many acceptable responses to this item and multiple different problems could be pointed out.